FBISE Math Class 11 Solved Model Paper New Pattern 2027

S No.	Question	A	B	C	D
1.	The linear factors of the polynomial $4z^2 + 9y^2$ are:	$(2z - 3yi)^2$	$(2z + 3yi)^2$	$(2z + 3y)(2z - 3y)$	$(2z + 3yi)(2z - 3yi)$

$z = \frac{-3iy}{2}$

$$4z^2 - (-9y^2)$$

$$= (2z)^2 - 9y^2 i^2$$

$$= (2z)^2 - (3yi)^2$$

$$= (2z + 3yi)(2z - 3yi)$$

$i^2 = -1$

$z = \frac{3iy}{2}$

$(2z - 3iy)$

$(2z + 3iy)$

2.	What is the principal argument of $-i - \sqrt{3}$?	$-\frac{4\pi}{3}$	$\frac{4\pi}{3}$	$-\frac{\pi}{3}$	$\frac{\pi}{3}$

$\frac{5\pi}{6}$

z

$$\boxed{-\pi \le \arg(z) \le \pi}$$

$-\frac{5\pi}{6}$

$z = -\sqrt{3} - i$

$\theta = \tan^{-1}\left(\frac{+1}{\sqrt{3}}\right)$

$= \frac{\pi}{6}$

ref. Angle

$\pi - \theta$ | θ

III $\theta - \pi$ | $-\theta$

$\text{Arg}(z) = \frac{\pi}{6} - \pi = \frac{\pi - 6\pi}{6}$

Prin. Arg(z) $= \frac{\pi}{6} - \pi = -\frac{5\pi}{6}$

A

3.	If $z = r\cos\theta + ir\sin\theta$ then z^{-1} is equal to:	$\frac{\cos\theta - i\sin\theta}{r}$	$\frac{-\cos\theta + i\sin\theta}{r}$	$\frac{-\cos\theta - i\sin\theta}{r}$	$\frac{\cos\theta + i\sin\theta}{r}$

→ Multiplicative Inverse

$z = x + yi$

$$\boxed{z^{-1} = \frac{1}{z} = \frac{x - yi}{x^2 + y^2}}$$

$\cos^2\theta + \sin^2\theta = 1$

$$z^{-1} = \frac{r\cos\theta - ir\sin\theta}{r^2\cos^2\theta + r^2\sin^2\theta}$$

$$z^{-1} = \frac{r\cos\theta - ir\sin\theta}{r^2(\cos^2\theta + \sin^2\theta)}$$

$$= \frac{\cancel{r}(\cos\theta - i\sin\theta)}{r^{\cancel{2}}}$$

$k^n|A|$

4.	If A is a square matrix of order 3 with $\lvert A\rvert = 5$, then the value of $\lvert 2A\rvert$ is:	5	10	25	40

$n = 3$ $\qquad |A| = 5$

$m \times n \to$ order n

$|2A| = ?$ $\qquad k \to$ const.

$(n) \times n$

$k \to$ const.

$|2A| = ?$

k

$$\boxed{|kA| = k^n |A|}$$

$$|2A| = 2^3 \times |A| = 8 \times 5 = 40$$

A $\quad -2$

5.	For $A = \begin{bmatrix} 3 & \text{[illegible]} & 1 \\ 0 & 5 & -1 \\ 2 & 1 & 4 \end{bmatrix}$, what is the value the cofactor A_{12}?	-2	0	2	-11

$$\underline{\underline{A_{12}}} = (-1)^{1+2} M_{12} = -(0-(-2))$$
$$= -2$$

$$M_{12} = \begin{vmatrix} 0 & -1 \\ 2 & 4 \end{vmatrix}$$

6.	Which condition defines an inconsistent system of linear equations?	One solution	Infinite solutions	Trivial solution	No solution

$x=0, y=0, z=0$

Consistent

Inconsistent

7.	If the first term of a sequence is 10 and the common difference is 0, what is the 50th term?	0	10	50	500

A.P

$d=0$ $\quad a_1, \; a_1+d, \; a_1+2d, \; \ldots, \; a_1+(n-1)d$

$a_1=10$ $\quad a_1, \; a_2, \; a_3, \; \ldots, \; a_n$

a_1 ,

$a_{50} = a_1 + 49d$

$a_{50} = a_1 = 10$

8.	Row 1 of a theatre has 20 seats. Each successive row has 2 additional seats. At which row number does the seat count reach 40?	10th	11th	12th	20th

$0 \to 10$

$0 \sim 9$ $a_1 = 20$, $d = 2$, $a_n = 40$

$1 \to 9$ 20, 22, 24, 26, ... , 40

$9 - 1 + 1$ Row 1 $n = ?$ a_n

$a_n = a_1 + (n-1)d$

$0 \sim 9$

$9 - 0 + 1$

$40 = 20 + (n-1)2$

$2(n-1) = 20$

$n - 1 = 10$

$n = 11$

$15 \to 71$

$71 - 15 + 1 =$ ✓

G.P

9.	A ball is dropped and rebounds to 60%, of its previous height. If the first bounce is 100 cm, what is the height of the 2^{nd} bounce?	60 cm ✓	48 cm	36 cm	6 cm

$n-1$

	what is the height of the 2^{nd} bounce?				

$a_1, a_1 r, a_1 r^2, \ldots, a_1 r^{n-1}$

$a_1 r^{n-1} \to a_n$

$r = \frac{60}{100}$

$r = 0.6$

r = common ratio

$a_1 = 100\ cm$

$r = 0.6$

$a_2 = ar = \frac{100 \times 0.6}{1} = 60\ cm$

10.	Find the 5^{th} term of an HP whose corresponding AP is 5, 10, 15, ...	$\frac{1}{20}$	$\frac{1}{25}$	$\frac{1}{30}$	$\frac{1}{35}$

A.P ⇒ 5, 10, 15, 20, 25

$\frac{1}{25}$

H.P

$a = 5, \ d = 5$

$a_5 = a + 4d$
$= 5 + 4(5) = 25$
↳ in A.P

a_5 in H.P $= \frac{1}{25}$

a_5 in HP $= \frac{1}{25}$

11.	The third term in the expansion of $(x^2-2)^4$ is:	$24\,x^4$	$-24\,x^4$	$-32\,x^2$	$32\,x^2$

$\binom{4}{0}(x^2)^4(-2)^0 + \binom{4}{1}(x^2)^3(-2)^1 + \binom{4}{2}(x^2)^2(-2)^2$

$\binom{4}{3}(x^2)^1(-2)^3$

$(a+b)^n$

General Term

$T_{r+1} = \binom{n}{r} a^{n-r} \cdot b^r$

$T_3 = T_{2+1} = \binom{4}{2}(x^2)^2(-2)^2$

$T_{3+1} = \binom{6}{3}(x^2)^3\left(-\frac{1}{x^2}\right)^3$

12.	Which term in the expansion of $\left(x^2 - \frac{1}{x^2}\right)^6$ is independent of x?	T_3	T_4	T_5	T_6

$x^0 \Rightarrow r = \frac{np}{p+q} = \frac{6\times 2}{2+2} = \frac{12}{4}$

$$\left(ax^p \pm \frac{1}{bx^q}\right)^n$$

$r = 3$

$T_{r+1} = T_4$

13.	The simplified form of $(x+1)^3 - (x-1)^3$ is:	$2x^3 + 6x$	$6x^2 + 2$	$6x^2 + 2x$	$2x^3 + 6x^2$

Let $x = 0.1$

$\frac{105}{50}$

$(x+1)^3 = x^3 + 1 + 3x^2 + 3x$

$(x-1)^3 = x^3 - 1 - 3x^2 + 3x$

14.	When 3^{50} is divided by 8, the remainder is:	1	3	5	7

$3^1 = 3 \equiv 3$

$3^2 = 9 = 8(1) + 1$

$3^2 = 9 \equiv 1$

$3^3 = 27 \equiv 3$

1 2 3, 4

3, 1, 3, 1

$(3^2)^{25}$

$= 1^{25}$

$= 1$

$9 \equiv 1 \pmod 8$

Remainder

$3^{50} \equiv 1$

15.	When dividing $x^3 + 2x^2 - 5x + 1$ by $x - 2$, the degree of the remainder is:	0	1	2	7

$x - 2 = 0$

$x = 2$

$x^2 \quad 5x^0$ Quot

$x - 2 \overline{\big) x^3 + 2x^2 - 5x + 1}$

16.	Dividing $x^4 - 3x^2 + 2$ by $x^2 - 1$, the quotient is:	$x^2 - 2$	$x^2 + 2$	$x^2 - 1$	$x^2 + 1$

$$
\begin{array}{r}
x^2 - 2 \\
x^2 - 1 \overline{\big) x^4 - 3x^2 + 2} \\
-\,(x^4 - x^2) \\
\hline
-2x^2 + 2 \\
-\,(-2x^2 + 2) \\
\hline
0
\end{array}
$$

17.	For what value of k, $x - 2$ is the factor of	-3	0	3	4

17.	For what value of k, $x-2$ is the factor of x^3-kx^2+4x-4 ?	-3	0	3	4

$x-2=0$

$x=2$

$p(x)=$

$p(2)=0$

$8-4k+8-4=0$

$\cancel{4}4k=\cancel{4}12$

$k=3$

18.	Two direction angles of vector $\vec{a}$ are 30° and 60°. What is the value of third direction angle?	135°	90°	45°	30°

$\alpha,\ \beta,\ \gamma$

$30^\circ \quad 60^\circ \quad ?$

$\boxed{\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1}$

$(\cos 30)^2+(\cos 60^\circ)^2+\cos^2\gamma=1$

$\frac{3}{4}+\frac{1}{4}+0=1$

$\cos^2\gamma=0$

$\cos\gamma=0$

$\gamma=\cos^{-1}(0)=90$

$v = \cos^{-1}(0) = (90)$

19.	The vector parallel to $2\hat{\imath} + 6\hat{\jmath} - 5\hat{k}$ and with magnitude 3 is:	$\left[\frac{12}{\sqrt{65}}, \frac{-18}{\sqrt{65}}, \frac{75}{\sqrt{65}}\right]$	$\left[\frac{12}{\sqrt{65}}, \frac{18}{\sqrt{65}}, \frac{25}{\sqrt{65}}\right]$	$\left[\frac{6}{\sqrt{65}}, \frac{18}{\sqrt{65}}, \frac{-15}{\sqrt{65}}\right]$	$\left[\frac{6}{\sqrt{65}}, \frac{-18}{\sqrt{65}}, \frac{8}{\sqrt{65}}\right]$

$\vec{V} = |\vec{V}|\,\hat{V}$

$= 3$

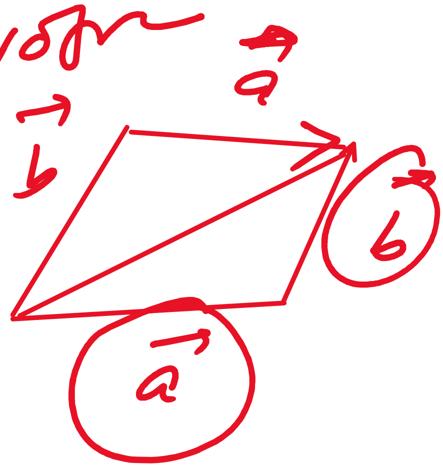
$\hat{V} = \frac{\vec{V}}{|\vec{V}|} = \frac{2\hat{i} + 6\hat{j} - 5\hat{k}}{\sqrt{65}}$

$(-5)^2$

$|\vec{V}| = \sqrt{4+36+25} =$

20.	The area of a parallelogram having adjacent sides $\vec{a} = [2, 5, -2]$ and $\vec{b} = [1, 4, -2]$ is:	$\sqrt{17}$	$\frac{\sqrt{17}}{2}$	$\frac{\sqrt{17}}{6}$	$2\sqrt{17}$

$|\vec{a} \times \vec{b}| =$ Area of ||gm



$|\vec{a} \times \vec{b}| = \begin{vmatrix} i & j & k \\ 2 & 5 & -2 \\ 1 & 4 & -2 \end{vmatrix}$

$= i(-10+8) - j(-4+2) + k(8-5)$

$= |-2i + 2j + 3k|$

$$= \frac{|-2i+2j+3k|}{\sqrt{4+4+9}}$$

$$= \sqrt{17}\ R$$

21.	At which angle the work done is zero?	0°	45°	90°	180°

$$W = Fd\cos\theta = 0$$

22.	In navigation the velocity of the air-plane is calculated using:	Dot product	Vector addition	Vector subtraction	Cross product

23.	If $\vec{a} = \hat{\imath} + \hat{2\jmath} - 2\hat{k}$ and $\vec{b} = \hat{2\imath} + 3\hat{\jmath} - 3\hat{k}$ then $\hat{\imath}.\vec{a} + \hat{\jmath}.\vec{b} = ?$	1	−2	2	4

$$\hat{i}\cdot(\hat{i}+2\hat{j}-2\hat{k}) = 1$$

$$\hat{i}\cdot\hat{j} = 0$$

$$\hat{i}\cdot\hat{k} = 0$$

$$\hat{j}\cdot\vec{b} = 3$$

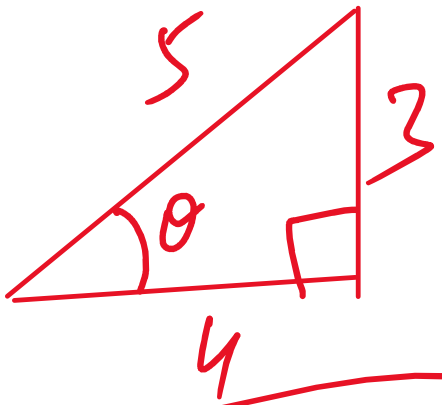
$$1+3 = 4$$

24.	If $\sin\theta = \frac{3}{5}$ and θ is acute, what is the value of $\sin 2\theta$?	$\frac{24}{25}$	$\frac{12}{25}$	$\frac{7}{25}$	$\frac{16}{25}$

$\sin 2(\theta)$

$3\text{-}4\text{-}5$ $5, 12, 13$

$\sin\theta = \frac{3}{5} \Rightarrow \theta = \sin^{-1}\left(\frac{3}{5}\right)$



P/H

$\sqrt{5^2 - 3^2} = \sqrt{25-9} = \sqrt{16} = 4$

$\cos\theta = \frac{4}{5}$

$\sin 2\theta = 2\sin\theta\cos\theta = \cancel{2\times\frac{3}{5}\times\frac{4}{5}} = \frac{24}{25}$

25.	If $\cos\theta = -\frac{1}{3}$ and θ lies in quadrant II, the value of $\sin\left(\frac{\theta}{2}\right)$ is:	$\frac{\sqrt{6}}{3}$	$-\frac{\sqrt{6}}{3}$	$\frac{\sqrt{3}}{3}$	$-\frac{\sqrt{3}}{3}$

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}}$

$$= \sqrt{1 + \frac{1}{\frac{3}{2}}}$$

26.	Simplifying $2(\sin 3x)(\cos x)$ into a sum or difference we get:	$\sin 4x + \sin 2x$	$\sin 4x - \sin 2x$	$\sin 2x - \sin x$	$\sin 2x + \sin x$

$$\boxed{2\sin\alpha\cos\beta = \sin(\alpha+\beta) + \sin(\alpha-\beta)}$$

$$= \boxed{\sin 4x + \sin 2x}$$

$$2\cos\alpha\sin\beta = \sin(\alpha+\beta) - \sin(\alpha-\beta)$$

27.	If $\tan\alpha = \frac{1}{2}$ and $\tan\beta = \frac{1}{3}$, (α, β are acute), the exact value of $(\alpha+\beta)$ is:	0°	8°	45°	90°

$$\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$$

$$= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2}\cdot\frac{1}{3}} = 1$$

calully $\tan(\alpha+\beta) = 1$ $\theta = 30^\circ$

28.	Which of the following is equivalent to $\sin(270^\circ - \theta)$?	$-\sin\theta$ ✗	$-\cos\theta$	$\sin\theta$ ✗	$\cos\theta$

Allied Angle

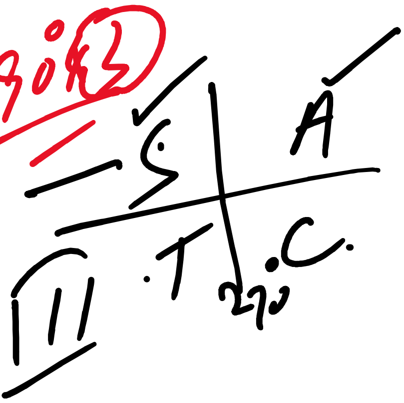
tan

$\sin(270^\circ - \theta) = -\cos\theta$

III

$270^\circ = 90^\circ \times 3$

odd multiple



29.	If $sin25^\circ = t$, then $cos50^\circ$ in terms of t is:	$2t^2 - 1$	$2t\sqrt{1-t^2}$	$1-2t^2$	$\sqrt{1-2t^2}$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$= 1 - \sin^2\theta - \sin^2\theta$$

$$\boxed{\cos 2\theta = 1 - 2\sin^2\theta}$$

$$\cos 50^\circ = 1 - 2t^2$$

$\sin 25^\circ = t$

$\sin y = 1$

30.	What is the range of the function $y = 3sin(2x)$?	$[-1,1]$	$[-2,2]$	$[-3,3]$	$(-\infty,\infty)$

$3\times [-1 \leq \sin(?) \leq 1]$

$[-3, +3]$

$-\sin x + \sin x = 0$

31.	The sum of $sin(-x) + sinx$ is?	0	1	$2sinx$	$sin2x$

$\sin(-x) = -\sin x \Rightarrow$ odd fn

$f(-x) = -f(x)$

$\cos(-x) = \cos x$

Even f

32.	The period of $4cos\left(\frac{x}{3}\right)$ is:	$\frac{4\pi}{3}$	$\frac{2\pi}{3}$	2π	6π

$\frac{1}{3}x$, period

$\frac{2\pi}{\frac{1}{3}} = 6\pi$

$\frac{x}{3} = 2\pi$

$x = 6\pi$

33.	What is the maximum value of $10\sin x - 5\cos x + 2$?	3	17	7	13

D

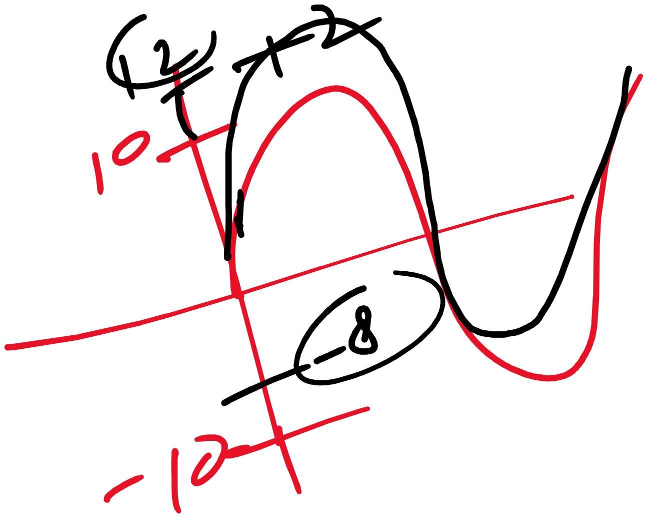
$$\text{Max. value} = \sqrt{(10)^2 + (-5)^2} + 2$$

$a\sin x \pm b\cos x$

$\text{Max Val} = \sqrt{a^2+b^2}$

$5\sqrt{5} + 2$

$= 13.18 \approx 13$



34.	Which pair of the trigonometric functions has period π?	$sinx, cosx$	$tanx, cotx$	$secx, cosecx$	$sinx, tanx$

35.	A pizza shop offers 8 toppings. How many different pizzas can be made if you must choose exactly 3 toppings?	24	56	92	336

order does not matter → Combination

$^8C_3$ ✓

36.	In how many ways can a student choose 4 questions out of 6 in an exam?	15	24	30	360

$^6C_4$ ✓ → Notes [illegible]

37.	In how many ways a DJ can play 5 songs out of 12 (order matters)? ✓	$\binom{12}{5}$	$\frac{12!}{5!}$	$^{12}P_5$	$\frac{12!}{4!}$

$$^{12}P_5 = \frac{12!}{7!}$$

$$^nC_r = \frac{n!}{r!\,(n-r)!}$$

38.	If a coin is tossed 3 times, what is the total number of possible outcomes?	3	6	8	9

1 coin {H, T}

No. of Tosses Total possible

1 coin = $\{H, T\}$

$2^{\text{No. of Toss}}$ = possible outcome

$2^3 = 8$

Die = $\{1, 2, 3, 4, 5, 6\}$

6^n

39.	The value of ${}^{n}P_{n-1}$ is?	n	$n!$	$\frac{n!}{(2n-1)!}$	$\frac{n}{2n-1}$

$$\frac{n!}{(n-(n-1))!} = \frac{n!}{(\cancel{n}-\cancel{n}+1)!} = n!$$

$${}^{n}P_{1} = \frac{n!}{(n-1)!} = \frac{n\cancel{(n-1)!}}{\cancel{(n-1)!}}$$

$${}^{n}P_{0}$$

$$\frac{n!}{n!} = 1$$

$n!.$

40.	If $^{n-1}P_3 : {}^{n}P_4 = 1:9$ then value of n is:	3	6	7	9

$$\frac{^{n-1}P_3}{^{n}P_4} = \frac{1}{9}$$

	STIMULUS/SITUATION/SCENARIO BASE MCQs				
	Scenario for MCQs (41-44)	**Given a complex polynomial $z^2 + 4$, where z is a complex number.**			
41.	The complex polynomial $z^2 + 4$ is:	reducible to real factors	irreducible to real factors	reducible to complex factors	irreducible to complex factors
42.	If 4 is replaced with 8 and z^2 is replaced with z^3 in polynomial $(z^2 + 4)$, then the number of linear factor(s) of polynomial are:	1	2	3	Not possible
43.	The linear factors of polynomial $z^2 + 4$ are:	$z + 4i$, $z - 4i$	$z - 2i$, $z - 2i$	$z + 2i$, $z - 4i$	$z + 2i$, $z - 2i$
44.	The two factors of $z^2 + 4$ are:	conjugate of each other	same	additive inverse of each other	reciprocal of each other

$z^3 + 8$

$z = 2i$

$z = -2i$

Scenario for MCQs (45-48)	**Three seismic stations track wave data where $x, y,$ and z represent the exact amplitude of primary, secondary, and surface waves respectively. The seismic network model is:** [1 1 1] [x] [6]

	Scenario for MCQs (45-48)	Three seismic stations track wave data where $x, y,$ and z represent the exact amplitude of primary, secondary, and surface waves respectively. The seismic network model is: $\mathbf{AX = B}$, $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 5 & 5 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 0 \end{bmatrix}$, then				
45.	The value of $\lvert A \rvert$ is:	-16	-8	0	16	○ ○ ○ ○
46.	The value of $\lvert A_z \rvert$ is:	-16	-60	0	60	○ ○ ○ ○
47.	The value of z is:	-3.75	0.27	3.75	7.5	○ ○ ● ○
48.	The value of $\lvert A \rvert$ if Row 3 is replaced by [3 3 3]	-48	-16	0	48	○ ○ ○ ○

$1*1$

$$\begin{bmatrix} \overset{+}{1} & \overset{-}{1} & \overset{+}{1} \\ 0 & 2 & 0 \\ 5 & 5 & 1 \end{bmatrix} \Rightarrow 2(1-5) = -8$$

$$= 2(-4)$$

$$|A_z| = \begin{vmatrix} 1 & 1 & 6 \\ 0 & 2 & 12 \\ 5 & 5 & 0 \end{vmatrix} = -60$$

$$Z = \frac{|A_z|}{|A|} = \frac{\cancel{-}60}{\cancel{-}8} = 7.5$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 3 & 3 & 3 \end{vmatrix}$$

$$1\ 1\ 1\ \ 0 = 0$$

$= 3$

Scenario for MCQs (49-52)

Use the table below showing the Weekly Savings of a student to answer the questions.

Week (n)	Total Savings (S_n)
1st	Rs 150
2nd	Rs 350
3rd	Rs 600
4th	Rs 900

49.	What is the value of a_2?	150	200	250	350	O O O O
50.	What is the total savings for the 3rd week?	200	250	350	600	O O O O
51.	Which formula represents the total savings (S_n) for any week (n)?	$s_n = \frac{n}{2}[300 + (n-1)50]$	$s_n = 150 + (n-1)50$	$s_n = \frac{n}{2}[150 + (n-1)50]$	$s_n = 150n + 50$	O O O O
52.	By the end of which week will the student's total savings first exceed Rs. 2,500?	6th week	7th week	8th week	9th week	O O O O

$S_n = \frac{n}{2}[300 + (n-1)50]$

S_1

$S_2 = a_1 + a_2 = 350$

$150, \quad 350$

$a_2 = 350 - 150$

$a_2 = 200$

$a_1, \quad a_2, \quad a_3 \ldots$

$S_1 \quad a_1, \quad a_2$

$150, \quad 200$

$d = 200 - 150$

$d = 50$

$$\boxed{S_n = \frac{n}{2}\left[2a_1 + (n-1)d\right]}$$

$$S_n = \frac{n}{2}\left[a_1 + a_n\right]$$

$$S_n = \frac{n}{2}\left[300 + (n-1)50\right]$$

	Scenario for MCQs (53-56)	A drone is flying in space. Its motion is represented by the vectors $\vec{a} = 2\hat{\imath} + \hat{\jmath} + 2\hat{k}$, $\vec{b} = \hat{\imath} + 2\hat{\jmath} + \hat{k}$. An engineer wants to analyze the relationship between these two directions using dot and cross products.				
53.	The engineer computes $\vec{a}\cdot\vec{b}$. What does this quantity help to determine?	The area of parallelogram formed by the vectors	The angle between the vectors	Law of sines	The area of a triangle	○ ○ ○ ○
54.	In terms of vectors $\vec{a}$ and $\vec{b}$, how would you express the work done on the drone?	$\|\vec{a}\| + \|\vec{b}\|$	$\|\vec{a} - \vec{b}\|$	$\vec{a} \times \vec{b}$	$\vec{a}.\vec{b}$	○ ○ ○ ○
55.	If $\vec{a}\times\vec{b} = 0$. What can be concluded about the drone's angle of direction?	0°	45°	60°	90°	○ ○ ○ ○
56.	An engineer wants to position a communication antenna perpendicular to both $\vec{a}$ and $\vec{b}$. Which vector gives its orientation?	$-6+6$ $-3\hat{\imath} + 3\hat{k}$ $-3+3$	$3\hat{\imath} + 3\hat{k}$	$-3\hat{\imath} - 3\hat{k}$	$6-6$ $3\hat{\imath} - 3\hat{k}$ $3+3$	○ ○ ○ ○

$$\vec{a}\cdot\vec{b} = ab\cos\theta$$

$$W = \vec{F}\cdot\vec{d} \qquad \cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}\vec{b}|}$$

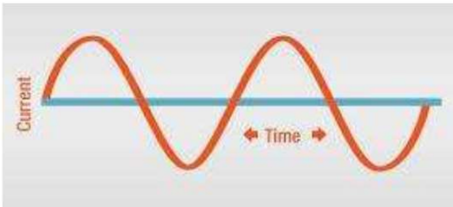
$$\vec{a}\times\vec{b} = ab\sin\theta\,\hat{n} = 0$$

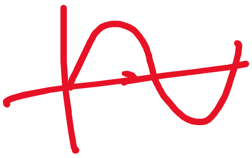
$\perp$ $\vec{a} \times \vec{b}$ is $\perp$ to both $\vec{a}$ & $\vec{b}$

$i \cdot j = 0$

$(\vec{a} \times \vec{b}) \cdot \vec{a} = 0$

$(\vec{a} \times \vec{b}) \cdot \vec{b} = 0$

	Scenario for MCQs (57-60) $-1 \le \sin x \le 1$	**Current in the circuit with the passage of time follows the wave pattern.** 			
57.	Wave shown in the figure is often known as:	secant wave	tangent wave	sine wave	longitudinal wave
58.	If the function $y = 2sinx$ represents the wave, then what is the domain of the function?	$(-\infty, 0]$	$[0, \infty)$	$(-\infty, \infty)$	$[1, \infty)$
59.	What is the maximum value of $y = 2sinx$?	0	1	2	∞
60.	If $y = 2sinx$ is replaced by $y = 3cosx$, then what is the range of the new function?	$(-3, 3)$	$[-3, 3]$	$(-3, 3]$	$[-3, 3)$



$3 \cos x$ $[-3, 3]$

	Scenario for MCQs (61-64)	**A student council has 12 members: 5 from Grade 11 and 7 from Grade 12. A committee of 4 members is to be formed. The committee must have exactly 2 students from Grade 11 and 2 from Grade 12. The order of selection does not matter.**				
61.	The number of ways to choose 2 students from Grade 11 is:	5	10	20	25	○ ○ ○ ○
62.	How many ways to choose 2 students from Grade 12?	7	14	21	42	○ ○ ○ ○

	2 students from Grade 11 is:					
62.	How many ways to choose 2 students from Grade 12?	7	14	21	42	○ ○ ○ ○
63.	The total number of ways to form the committee is:	31	210	420	840	○ ○ ○ ○
64.	If the committee instead had no grade restriction, the number of possible committees would be:	210	330	495	792	○ ○ ○ ○

Total 12

Grade 11 5 → 2

Grade 12 7 → 2

Committee member = 4

61) $^{5}C_{2} = 10$

62) $^{7}C_{2} = 21$

63) $^{5}C_{2} \times {}^{7}C_{2} = 10 \times 21 = 210$

Total

64) $^{12}C_{4} = 495$